

EID/Email:

1. [9 pts] Define the von Neumann's architecture. Include both how a von Neumann machine is wired and the process used to execute programs.
2. [9 pts] Define the von Neumann Bottleneck and explain how it is related to the Processor-Memory performance gap.
3. [9 pts] List the computational paradigms discussed *in class*.

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4. [9 pts] How are compilers and interpreters similar? How are they different?
5. [9 pts] What two language design criteria focus areas would you prioritize when designing a programming language? Why?
6. [9 pts] Discuss two or more examples of efficiency in the Java programming language. Specify if your example is computational efficiency or programmer efficiency.

7. [9 pts] Discuss two or more examples of inefficiency in the Java programming language. Specify if your example is computational efficiency or programmer efficiency.

8. [9 pts] Discuss two or more examples of regularity in the Java programming language.

9. [9 pts] Discuss two or more examples of irregularity in the Java programming language.

10. [10 pts] Should a language require the declaration of variables? Languages such as Lisp and Python allow variables names to be used without declarations, while C, Java, and Ada require all variables to be declared. Discuss the requirement that variables should be declared from the point of view of readability, writability, efficiency, and security.

11. [9 pts] List the two types of programming language abstractions and the three levels of programming language abstractions